

GAUTAM FruitSense

AI-Powered Post-Harvest Fruit Quality Intelligence

Comprehensive Project Report

Problem Statement | Solution | Model Architecture

Dataset Analysis | Training Workflow | Real-World Implementation

98.03%

Test Accuracy

18,000+

Images

10

Classes

MobileNetV2

Architecture

Date: February 12, 2026

Team: GAUTAM Project

Technology: TensorFlow + React + FastAPI

1. Problem Statement

Post-harvest food loss is one of the most pressing global challenges. According to the Food and Agriculture Organization (FAO), approximately **1.3 billion tonnes of food** are wasted annually worldwide. In developing countries, **40-50%** of fruits and vegetables are lost after harvest due to inadequate quality assessment, improper storage, and lack of real-time monitoring systems.

Key Challenges:

- **Subjective Inspection:** Manual quality grading by human inspectors is inconsistent, error-prone, and impossible to scale.
- **No Real-Time Assessment:** There is no accessible, low-cost system for farmers, retailers, or consumers to instantly assess fruit quality.
- **Shelf Life Uncertainty:** Without objective data, estimating remaining shelf life relies on guesswork, leading to premature disposal or unsafe consumption.
- **Storage Guidance Gap:** Generic storage advice does not account for the specific fruit type, its current condition, or environmental factors.
- **Supply Chain Opacity:** Lack of standardized quality data across the supply chain prevents optimal routing, pricing, and waste reduction.

This project addresses these challenges by developing **GAUTAM FruitSense** — an AI-powered system that uses deep learning to provide instant, objective fruit quality assessment with actionable insights.

2. Proposed Solution

GAUTAM FruitSense is an end-to-end AI system comprising three integrated layers:

Layer	Technology	Function
Deep Learning Engine	TensorFlow / Keras / MobileNetV2	Classifies fruit images into 10 categories (5 fruits x Fresh/Rotten)
Backend API	FastAPI + Uvicorn	Accepts image uploads, runs inference, returns quality metrics via JSON
Frontend Dashboard	React 19 + Vite	Upload-centric UI with interactive results, charts, AI chatbot, and PDF reports

Core Capabilities:

- Real-time fruit classification (Apple, Mango, Orange, Potato, Tomato) as Fresh or Rotten
- Smart quality scoring (0-100) using blended probability analysis, not just binary classification
- Estimated shelf life, ripeness, freshness, and defect count per fruit
- Dynamic storage guide: different advice for fresh vs. rotten fruit (fridge, freezer, room temp)
- Chef-level culinary data: recipes, cheese/spice pairings, flavor profiles, and varieties
- Comprehensive A-to-Z PDF reports for supply chain documentation
- AI chatbot for interactive fruit quality Q&A;

3. Model Architecture & Algorithms

3.1 Base Model: MobileNetV2

MobileNetV2 is a convolutional neural network architecture optimized for mobile and embedded devices. It uses **inverted residual blocks** with **depthwise separable convolutions**, reducing parameters by 10-40x compared to traditional CNNs (e.g., VGG16: 138M params vs MobileNetV2: 3.4M params) while maintaining comparable accuracy.

Property	Value
Architecture	MobileNetV2 (pre-trained on ImageNet, 1.4M images)
Key Innovation	Depthwise separable convolutions + linear bottlenecks
Total Parameters	~4.38M (with custom head)
Input Shape	224 x 224 x 3 (RGB)
Output Classes	10 (5 fruits x Fresh/Rotten)
Activation	ReLU6 (base) + Softmax (output)

3.2 Custom Classification Head

The pre-trained MobileNetV2 base is connected to a custom classification head designed for our specific 10-class problem:

Layer	Output Shape	Parameters	Notes
GlobalAveragePooling2D	1280	0	Spatial reduction
BatchNormalization	1280	5,120	Stabilize training
Dense + ReLU	512	655,872	L2 regularization (0.001)
Dropout	512	0	Rate: 0.5 — prevents overfitting
Dense + ReLU	256	131,328	L2 regularization (0.001)
Dropout	256	0	Rate: 0.4
Dense + ReLU	128	32,896	Feature refinement
Dropout	128	0	Rate: 0.3
Dense + Softmax	10	1,290	Final classification

3.3 Transfer Learning Strategy

Instead of training from scratch, we leverage transfer learning: the base MobileNetV2 retains knowledge from 1.4M ImageNet images, and we fine-tune it for fruit quality classification. This dramatically reduces training time and improves accuracy with limited data.

4. Training Workflow

4.1 Three-Phase Progressive Fine-Tuning

We use a progressive unfreezing strategy across 3 training phases for maximum accuracy:

Phase	Strategy	Epochs	LR	Layers Unfrozen
Phase 1	Train classification head only	15	0.001	Head only (~826K)
Phase 2	Fine-tune top 50 layers	20	0.0001	Top 50 + head
Phase 3	Full fine-tune (all layers)	15	0.00002	All 155 layers

4.2 Data Augmentation Pipeline

Heavy augmentation is applied only to training images to make the model robust to real-world conditions:

Augmentation	Range	Purpose
Random Rotation	+/- 40 degrees	Handle any fruit orientation
Width & Height Shift	+/- 30%	Handle off-center placement
Shear Transform	20%	Simulate perspective distortion
Zoom	+/- 30%	Handle varying camera distances
Horizontal Flip	Yes	Mirror invariance
Brightness Variation	0.7x to 1.3x	Handle different lighting
Channel Shift	+/- 30	Handle color temperature changes

4.3 Optimizer & Callbacks

- **Optimizer:** Adam with phase-specific learning rates (0.001 -> 0.0001 -> 0.00002)
- **Loss Function:** Categorical Cross-Entropy (multi-class classification)
- **ModelCheckpoint:** Saves best model by validation accuracy
- **EarlyStopping:** Patience 5-8 epochs; restores best weights
- **ReduceLROnPlateau:** Halves LR when validation loss plateaus for 2-3 epochs
- **Regularization:** L2 weight decay (0.001) + Dropout (0.3-0.5)

5. Dataset Details

5.1 Visual Dataset (Image Classification)

Split	Purpose	Approx. Images
Train	Model training with augmentation	~12,600
Validation	Hyperparameter tuning & early stopping	~3,600
Test	Final evaluation on unseen data	~1,800
Total		~18,000

5.2 Class Distribution

Class	Fruit	State	Data Sources
FreshApple	Apple	Fresh	Visual images + Tactile sensor data
RottenApple	Apple	Rotten	Visual images + Tactile sensor data
FreshMango	Mango	Fresh	Visual images + Tactile sensor data
RottenMango	Mango	Rotten	Visual images + Tactile sensor data
FreshOrange	Orange	Fresh	Visual images + Tactile sensor data
RottenOrange	Orange	Rotten	Visual images + Tactile sensor data
FreshPotato	Potato	Fresh	Visual images + Tactile sensor data
RottenPotato	Potato	Rotten	Visual images + Tactile sensor data
FreshTomato	Tomato	Fresh	Visual images + Tactile sensor data
RottenTomato	Tomato	Rotten	Visual images + Tactile sensor data

5.3 Tactile / Compression Dataset

In addition to visual data, the dataset includes **~5,987 tactile sensor readings** capturing force-displacement curves from compression tests. Fresh fruits exhibit higher compression resistance (firmer texture), while rotten fruits show lower force values (softer, decomposing tissue).

Property	Value
Total Sensor Files	~5,987 (Fresh: 2,995 Rotten: 2,992)
Data Type	Force-displacement curves (kN vs mm)
Purpose	Mechanical property analysis for multi-modal quality assessment

Property	Value
Additional Files	fresh_dimensions.ods, rotten_dimensions.ods (physical measurements)

6. Smart Quality Scoring Algorithm

Unlike traditional binary classification that outputs only "Fresh" or "Rotten", our system computes a **blended quality score (0-100)** by analyzing ALL class probabilities from the softmax output. This gives nuanced scores: a half-rotten fruit gets ~40, not 0.

Algorithm:

- 1. Run image through MobileNetV2 -> Get 10-class softmax probabilities
- 2. Identify the top predicted class (e.g., RottenTomato at 85%)
- 3. Extract both fresh and rotten probabilities for the detected fruit type
- 4. Compute $\text{freshness_ratio} = \text{FreshProb} / (\text{FreshProb} + \text{RottenProb})$
- 5. If Fresh: $\text{score} = 60 + \text{freshness_ratio} * 38$ (range: 60-98)
- 6. If Rotten: $\text{score} = 15 + (1 - \text{confidence}) * 30 + \text{freshness_ratio} * 15$ (range: 10-45)
- 7. Derive sub-metrics: ripeness, freshness %, defect count, shelf life from score

Condition	Score Range	Label	Action
100% Fresh (high confidence)	85-98	Excellent	Store normally, consume within 5-7 days
Fresh (moderate confidence)	60-85	Good / Fair	Use within 3-5 days, monitor closely
Half Rotten (partial decay)	30-45	Moderate Decay	Cut away bad parts, use immediately
Mostly Rotten	15-30	Significant Decay	Inspect carefully, consider discarding
Fully Rotten	10-15	Severely Deteriorated	Discard immediately, compost

7. Results & Performance

Metric	Value
Test Accuracy	98.03%
Best Validation Accuracy	98.49%
Test Loss	0.0790
Epochs Trained	28
Model Size	~15 MB (.h5 format)
Inference Time (CPU)	~500ms per image
Inference Time (GPU, projected)	<50ms per image

The model achieves **98.03% test accuracy** on unseen data, demonstrating strong generalization. The 3-phase progressive fine-tuning strategy contributed to consistent improvement across all phases, with early stopping preventing overfitting.

8. System Features

- **Upload-Centric Analysis:** All detailed information appears ONLY after uploading an image. Clean initial UI with hero section and upload zone.
- **Freshness-Aware Results:** Fresh fruit shows green fridge guide (whole, cut, freezer durations). Rotten fruit shows red WARNING section with "Discard Immediately" cards and spoilage indicators.
- **Smart Quality Scoring:** Uses all 10 softmax probabilities for blended scoring (0-100). Half-rotten fruit gets ~40, not zero.
- **Chef's Corner:** Recipes, cheese/spice pairings, flavor profiles, culinary uses, and popular varieties — all specific to the detected fruit.
- **Dynamic Fridge Guide:** Specific storage durations for whole in fridge, cut in fridge, room temperature, and freezer — tailored to each fruit type.
- **PDF Report Export:** 10-section professional report with dark cover page, quality metrics, nutrition, culinary guide, compression data, dataset info, and recommendations.
- **AI Chatbot:** Interactive chatbot answering questions about fruit storage, nutrition, recipes, and model architecture details.
- **Visual Analytics:** Radar chart (quality breakdown), line chart (quality trajectory), and bar chart (defect analysis) using Recharts.

9. Real-World Implementation Plan

The GAUTAM FruitSense system is designed for progressive real-world deployment across multiple use cases and environments:

Phase 1: Supermarket Quality Stations

Deploy touchscreen kiosks at produce sections in supermarkets and grocery stores. Customers photograph fruit using a mounted camera. The system instantly displays a quality score, estimated shelf life, and storage recommendations. This empowers consumers to make informed purchasing decisions and reduces returns due to unexpected spoilage.

Phase 2: Packaging & Sorting Facility Integration

Install industrial cameras on conveyor belts at fruit packaging and sorting facilities. The system automatically grades each fruit into quality tiers (Grade A/B/C) based on visual analysis. This replaces subjective manual inspection with consistent, high-throughput automated grading. Output data feeds into ERP systems for dynamic pricing and routing.

Phase 3: Mobile App for Farmers & Distributors

Develop a lightweight mobile application (Android/iOS) that allows farmers to photograph harvested produce directly in the field. The AI model runs on-device (using TensorFlow Lite) for instant quality assessment without internet connectivity. This helps farmers decide optimal harvest timing and identify the best market window for their produce.

Phase 4: IoT + Cold Chain Monitoring

Integrate the visual quality assessment system with IoT temperature/humidity sensors installed in refrigerated trucks and cold storage facilities. By combining real-time environmental data with periodic image-based quality checks, the system generates predictive spoilage alerts and recommends route/destination adjustments to minimize waste across the entire supply chain.

Scalability Roadmap

Metric	Current (Prototype)	Production Target
Fruits Supported	5 types (10 classes)	50+ types (100+ classes)
Inference Speed	~500ms (CPU)	<50ms (GPU / TF Lite)
Throughput	1 image/sec	50+ images/sec (batch GPU)
Deployment	Local development server	Docker + Kubernetes + CDN
Data Sources	Visual images only (inference)	Visual + Tactile + IoT sensors
Model Updates	Manual retraining	Automated ML pipeline (MLOps)

10. Technology Stack

Layer	Technology	Version / Details
Deep Learning	TensorFlow + Keras	2.x with MobileNetV2 transfer learning
Backend API	FastAPI + Uvicorn	Async REST API with auto-reload development server
Frontend	React + Vite	React 19 with Vite 7 build system
Charts	Recharts	Radar, Line, and Bar charts for visual analytics
PDF Generation	jsPDF + jspdf-autotable	Client-side A-to-Z report generation
Icons	Lucide React	Modern SVG icon library
Image Processing	Pillow + OpenCV	Image loading, resizing, and preprocessing
Numerical	NumPy	Array operations for model input/output
Styling	Vanilla CSS	Dark theme + glassmorphism + responsive design
Language	Python 3.11 + JavaScript ES2023	Backend/ML in Python, Frontend in JS

11. Conclusion

GAUTAM FruitSense demonstrates that deep learning can be effectively applied to post-harvest fruit quality assessment with high accuracy (98.03%) and practical utility. The system goes beyond simple classification by providing actionable insights — shelf life estimation, storage guidance, culinary recommendations, and comprehensive reporting.

The 3-phase progressive fine-tuning strategy, combined with heavy data augmentation and smart probability-based scoring, ensures robust performance across varying lighting conditions, orientations, and decay stages. The upload-centric frontend design ensures that all information is contextually relevant to the user's specific fruit.

With a clear roadmap for real-world deployment — from supermarket kiosks to mobile apps to IoT-integrated cold chain monitoring — GAUTAM FruitSense has the potential to make a meaningful impact in reducing the estimated 1.3 billion tonnes of annual food waste worldwide.